SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted(Yes/ N0/ NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example similar)	NA
6	Deployed application URL (if hosted)	No
7	APK/Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	No

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below

```
flood_prediction/
├── dataset/                      # Folder containing datasets
│   ├── train.csv                 # Training dataset from Kaggle
│   └── test.csv                  # Testing dataset from Kaggle
├── model/                        # Folder for storing trained models
│   └── flood_model.pkl           # Saved XGBoost trained model
├── templates/                    # HTML templates for Flask application
│   ├── index.html                # Home page with input form
│   ├── result.html               # Prediction result page
│   └── about.html                # About page with project details
├── static/                       # Static files used by the website
│   ├── css/
│   │   └── style.css             # Stylesheet for webpage design
│   └── images/                   # Images used in the application
├── app.py                        # Main Flask web application
├── train_model.py                # Script to train the ML model
├── requirements.txt              # Python dependencies/libraries
├── Procfile                      # Deployment configuration file
├── runtime.txt                   # Python version specification
└── README.md                     # Project documentation and setup guide
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://chetanreddt.pythonanywhere.com/
Login credentials(Demo)	
Platform/Hosting Provider	Local machine using Flask development server
Repository Link	https://github.com/yasminshaik18/Flood_prediction.git
Demo Video Link	

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

Step 1 — Clone or download the project folder to your computer

Step 2 — Open terminal or command prompt and navigate to the project folder:cd flood_prediction

Step 3 — Install all required Python libraries:pip install -r requirements.txt

Step 4 — Train the machine learning model:python train_model.py(This will create flood_model.pkl in the model folder)

Step 5 — Start the Flask web application:python app.py

Step 6 — Open your browser and go to:http://127.0.0.1:5000

Step 7 — Enter values from 1 to 10 for all 20 environmentalparameters in the form

Step 8 — Click "Analyze Flood Risk Now" button to get the prediction result

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Work around / Status
1	Model trained on only 50,000 rows out of 1.1 million records due to hardware limitations	Accuracy is 81.60% which is acceptable for academic use. Full dataset training possible on higher spec machine
2	Application is not deployed on a live cloud server	Can be run locally by following the setup instructions provided above
3	No real-time weather data integration — user must manually enter all input values	Future enhancement planned to integrate weather API for automatic input population